

K- Theory

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1 G -bundles over G spaces

G is a topological group, then by G -space we mean a topological space X together with a continuous action of G on X . A G -map is a map that respects the G -action between G -spaces.

A G -space E is a G -vector bundle over the G -space X if

1. E is a vector bundle over X .
2. the projection $E \rightarrow X$ is a G -map.
3. for each $g \in G$ the map $E_x \rightarrow E_{gx}$ is a vector space homomorphism.

If $X = \{x\}$ then a vector bundle over X is just a representation space of G over the fiber vector space.

G is a finite group. Suppose X is a free(faithful) G -space. Let X/G be the orbit space. If E is a G vector bundle over the space X , then E is also a free G space. and E/G has a natural vector bundle structure over X/G . Conversely if F is a vector bundle over X/G , then $q^*(F) \rightarrow X$ is a G -equivariant bundle map and $q^*(F)_x \simeq q^*(F)_{gx}$.

Proposition 1.1. *If X is a free G , then*

$$\{G - \text{vector bundles over } X\} \leftrightarrow \{\text{vector bundles over } \frac{X}{G}\}$$

by the map $E \rightarrow E/G$.

The natural map for any V ,

$$\sum V_i \otimes \text{hom}_G(V_i, V) \rightarrow V$$

is a G -isomorphism.

Define for a E a G -bundle over a trivial G -space X , $Av \in \text{END}(V)$ by

$$Av(e) = \frac{1}{|G|} \sum_{g \in G} g(e)$$

$\dim(\text{hom}^g(V, W))$
is the
multiplicity of
 W in V .

This is a projection operator and the image is a vector subbundle denoted by E^G , the invariant sub-bundle of E . So if E, F are two G bundles then $\text{hom}_G(E, F) = \text{hom}(E, F)^G$ is again a vector bundle.

Taking E to be the trivial vector bundle $V_i = X \times V_i$ we have a natural bundle map which is an isomorphism of G -bundles

$$\sum_{i=1}^k V_i \otimes \text{hom}_G(V_i, F) \rightarrow F$$

Thus every G -bundle G is isomorphic to a G -bundle of the form $\sum V_i \otimes E_i$ where E_i is a vector bundle with trivial G -action, which is unique upto isomorphism ($\simeq \text{hom}_G(V_i, F)$).

Proposition 1.2. *Let X be a trivial G -space, and $V_1, \dots, V_k$ be a complete set of irreducible G -modules, which are trivial over X . Then every G -bundle F over X is isomorphic to a direct sum $\sum V_i \otimes E_i$, where the $E_i \simeq \text{hom}_G(V_i, F)$ are unique upto isomorphism with trivial G -action.*

If E is a G -bundle then we have

$$\begin{aligned} G \times \Gamma(E) &\rightarrow \Gamma(E) \\ g, s &\mapsto g s g^{-1} \end{aligned}$$

The operator $Av = \frac{1}{|G|} \sum_{g \in G} g$ is a projection operator from $\Gamma(E) \rightarrow \Gamma(E)^G$.

Lemma 1.3. *Let X be a compact G -space and $Y \subset X$ be a closed sub G -space. Let E be a G -bundle over X . Then any invariant section s over Y extends to an invariant section over X .*

Proof. Extend s to a section t over X . Then $Av(t)$ is an invariant section of E over X which is equal to s over Y . $\square$

If E, F are two G -bundles then $\text{hom}(E, F)$ is also G -bundle and we have $\Gamma(\text{hom}(E, F))^G \simeq \text{HOM}_G(E, F)$

Lemma 1.4. *Let Y be a compact G space and X be a G -space. $f_t : Y \rightarrow X$, G homotopy and E a G vector bundle over X . Then $f_0^*(E)$ and $f_1^*(E)$ are isomorphic G bundles*

A G -homotopy means a G -map $F : Y \times I \rightarrow X$ where I is the unit interval with trivial G action (which is $f_t(g \cdot y) = g f_t(y)$). A trivial G -bundle we mean a G -bundle isomorphic to a product $X \times V$ where V is a G -module. h is a metric for E implies $Av(h)$ is an invariant metric.

If $V \subset \Gamma(E)$ is ample then $\sum gV \subset \Gamma(E)$ is ample and invariant.

To classify G -bundles, we look at Grassmanians of G -subspaces of $m \sum_{i=1}^k V_i = m[G]$ for $m \rightarrow \infty$.

$$\lim_{m \rightarrow \infty} [X, G_n(mR[G])]^G \simeq G \text{Vect}_n(X)$$

$A = C(X)$. If X is a G -space, G acts on A as a group of algebra automorphisms.

If E is a G -vector bundle over X , then $\Gamma(E)$ is a projective A -module, and G acts on $\Gamma(E)$,

$$g(as) = g(a)g(s) \quad a \in A, g \in G, s \in \Gamma(E)$$

Consider the twisted group algebra B of G over A , elements of B are linear combinations $\sum_{g \in G} a_g g$ with $a_g \in A$ and the products is

$$(ag)(a'g') = (ag(a'))gg'$$

Then $\Gamma(E)$ is a B -module. So Γ is a functor from G -vector bundles to finitely generated B -modules. If E is a trivial G -bundle, then $\Gamma(E)$ is free of rank n .

So Γ is now a functor inducing equivalences between category of trivial vector bundles and category of free B -modules. Consider the projection operator on these two categories. This gives us an equivalence

$$\{G\text{-vector bundles}\} \leftrightarrow \{\text{finitely generated } B\text{-modules and projective over } A\}$$

2 Periodicity theorem

E vector bundle over the space X , and if $E_0 = E - X$, X is considered to lie in E as the zero section. The group $\mathbb{C}^*$ acts on E_0 as the group of fiber preserving automorphisms. Let $P(E) = E_0 / \sim$, this is projective bundle associated to E .

Let $p : P(E) \rightarrow X$ be the projection map. So $p^{-1}(x)$ is a $\mathbb{C}$ -vector space for each x . If V is a vector space and W is a one dimensional vector space, then $V \cong V \otimes W$ $v \mapsto v \otimes w$, but not naturally. This defines an isomorphism, $P(w) : P(V) \rightarrow P(V \otimes W)$. For any other nonzero element $w' \in W$, we have $w' = \lambda w$. So $P(w) = P(w')$ and the isomorphism between $P(V) \cong P(V \otimes W)$ is natural. Thus for E vector bundle and L line bundle, we have

$$P(E) \cong P(E \otimes L)$$

If E is a vector bundle over X , then each point $a \in P(E)_x = P(E_x)$ represents a one dimensional subspace $H_x^* \subset E_x$. Then

$$H^* = \bigcup_{x \in X} H_x^* \subset p^*E$$

is a subbundle, where $p : P(E) \rightarrow X$ is the projection map. Since the problem is local we may assume $E \cong X \times V$ and we have $p(E)_x = \{x\} \times Gr_1 V$.

H^* is used because we want dual H , choice from algebro-geometric considerations

Theorem 2.1 (Periodicity theorem). *Let L be a line bundle over X . Then as a $K(X)$ algebra,*

$$K(P(L \oplus 1)) = \frac{K(X)[H]}{([H] - [1])([L][H] - [1])}$$

Note that $P(1 \oplus 1) = X \times S^2$.

Corollary 2.2.

$$K(S^2) = \frac{K(\text{pt.})[H]}{([H] - 1)^2} = \frac{\mathbb{Z}[H]}{([H] - 1)^2}$$

Corollary 2.3. *If X is any space and if*

$$\begin{aligned} \mu : K(X) \otimes K(S^2) &\xrightarrow{\cong} K(X \times S^2) \\ a \otimes b &\mapsto \pi_1^*(a) \otimes \pi_2^*(b) \end{aligned}$$

where π_i are the projections onto the two factors, is an isomorphism of rings.

For any $x \in X$ there is a natural embedding

$$\begin{aligned} L_x &\rightarrow P(L \oplus 1)_x \\ y &\mapsto (y, 1) \end{aligned}$$

This map extends to the one point compactification of L_x onto $P(L \oplus 1)_x$

$$\begin{aligned} L_x^+ &\xrightarrow{\cong} P(L \oplus 1)_x \\ \{\infty\} &\mapsto (1, 0) \end{aligned}$$

If we map

$$\begin{aligned} s_\infty : X &\rightarrow P(L \oplus 1) \\ x &\mapsto (1, 0) \end{aligned}$$

we obtain a section of $P(L \oplus 1)$, called the section of infinity.

Choose a metric for L . Let $S \subset L$ be the S^1 bundle.

$$P^0 = \{v \mid |v| \leq 1, v \in L\} \quad P^\infty = s_\infty(X) \bigcup \{v \mid |v| > 1, v \in L\}$$

We denote the projections as

$$\begin{array}{ccc} S & P^0 & P^\infty \\ \downarrow \pi & \downarrow \pi_0 & \downarrow \pi_\infty \\ X & X & X \end{array}$$

Since π_0 and π_∞ are homotopy equivalences, every bundle on P^0 is of the form $\pi_0^* E^0$ and every bundle on P^∞ is of the form $\pi_\infty^* E^\infty$, where E^0 and E^∞ are bundles on X . Thus, any bundle E on $P(L \oplus 1)$ is isomorphic to one of the form $\pi_0^*(E^0) \bigsqcup_f \pi_\infty^*(E^\infty)$ where $f \in \text{Iso}(\pi^*(E^0), \pi^*(E^\infty))$ is a clutching function.

We look to approximate any gluing function using Laurent clutching functions,

$$f = \sum_{k=-n}^{k=n} a_k z^k$$

These finite series can be used to approximate functions on S in a uniform manner.

If L is trivial, then $S = S \times S^1$. and the projection $z : S \rightarrow S^1$ is just the projection map.

If $L = 1$, we have $P = \mathbb{C}P^1 \times X$ and given bundles E_0 and E_∞ over X , any choice of a section $a : X \rightarrow \text{Iso}(E_\infty, E_0)$ gives a gluing function

$$S = S^1 \times X \ni (z, x) \xrightarrow{f} a(z) \in \text{Iso}(\pi^*(E_\infty), \pi^*(E_0))_{(z,x)}$$

To carry over the definition to a general line bundle $L \rightarrow X$ we define z as the section

$$S \ni [1 : z] \mapsto ([1 : z], z) \in \pi^* L$$

We want the gluing function f to be a section of the bundle $\text{Iso}(\pi^*(E_\infty), \pi^*(E_0))$. We take coefficients a to be a section $X \rightarrow \text{Iso}(E_\infty, L \otimes E_0)$ so that the formula

$$S \ni [1 : z] \xrightarrow{f} f z^{-1} a(\pi([1 : z])) \in \text{Iso}(\pi^*(E_\infty), \pi^* L^{-1} \otimes \pi^*(L \otimes E_0)) = \text{Iso}(\pi^*(E_\infty), \pi^*(E_0))$$

makes sense.

We write L^{-k} for $\pi^*(L^k)$ and we have $L^k \otimes L^{k'} \cong L^{k+k'} \quad \forall k, k'$.

We let $L \rightarrow X$ be an arbitrary line bundle and choose $E_0 = 1$ and $E_\infty = L$ and

$$z^{-1} \in \Gamma(\pi^*(L^{-1})) = \text{Iso}(1, \pi^* L^{-1}) = \text{Iso}(\pi^* L, 1) = \text{Iso}(\pi^* E_\infty, \pi^*(E_0))$$

a suitable gluing function. The resulting line bundle $\mathcal{V}(L, z^{-1}, 1) \simeq H$ is the canonical bundle over P .

More generally for arbitrary powers of z and index $k \in \mathbb{Z}$, a_k , we have the setup:
Given an exponent $k \in \mathbb{Z}$ and any section $a_k \in \Gamma\text{Hom}(L^k \otimes E_\infty, E_0)$ the formula

$$S \ni [1 : z] \xrightarrow{f} z^k a_k(\pi([1 : z])) \in \text{Iso}(\pi^*(L^k \otimes E_\infty), \pi^*L^k \otimes \pi^*E_0) = \text{Hom}(\pi_*E_\infty, \pi^*(E_0))$$

defines a section $a_k z^k \in \Gamma\text{Hom}(\pi^*E_\infty, \pi^*E_0)$ and we have the relation

$$\mathcal{V}(E_\infty, z^k a_k, E_0) \simeq H^k \otimes \mathcal{V}(L^k \otimes E_\infty, a_k, E_0)$$

Suppose that $f \in \Gamma\text{Hom}(\pi^*E_\infty, \pi^*E_0)$, we define it's Fourier coefficients $a_k \in \Gamma\text{Hom}(L^k \otimes E_0, E_\infty)$ by

$$a_k(x) = \frac{1}{2\pi i} \int_{S_x} f_x z_x^{-k-1} dz_x$$

where f_x, z_x denote the restrictions of f, z to S_x and dz_x is therefore a differential on S_x with coefficients in L_x . Let $S_n = \sum_{-n}^n a_k z^k$ and define the Cesaro means $f_n = \frac{1}{n} \sum_{k=0}^{n-1} S_k$. Then the Fejer's theorem on the $C^0(S^1)$ summability of Fourier series extends to vector bundles and gives

Proposition 2.4. *Let f be any clutching function for $\mathcal{V}(E_0, E_\infty)$ and let f_n be the sequence of Cesaro means of the Fourier series of f . Then f_n converges uniformly to f . Thus for large n , f_n is a clutching function for $\mathcal{V}(E_0, E_\infty)$ and*

$$\mathcal{V}(E_0, f, E_\infty) \cong \mathcal{V}(E_0, f_n, E_\infty)$$

We also get for large n , $0 \leq t \leq 1$, $|t f + (1 - t) f_n| \in \text{Iso}(E_0, E_\infty)$.

We now look at gluing functions that are polynomial in z . Let E_0 and E_∞ be vector bundles over X and let for some $n \in \mathbb{N}$ $p(z) = \sum_{k=0}^n a_k z^k$ be a polynomial gluing function of degree at most n . We introduce the auxiliary bundle $E_n^0 = \bigoplus_{k=1}^n L^k \otimes E_0$. Define a linearization of p by defining the section

$$\text{lin}_n p \in \Gamma\text{Hom}((\pi^*(\bigoplus_{k=0}^n L^k \otimes E_0)), \pi^*(E_\infty \oplus \bigoplus_{k=1}^n L^k \otimes E_0))$$

by the matrix

$$\text{lin}_n p(z) = \begin{pmatrix} a_0 & a_1 & a_2 & \dots & a_n \\ -z & 1 & & & \\ & -z & 1 & & \\ & & \ddots & \ddots & \\ & & & -z & 1 \end{pmatrix}$$

It has properties $\text{lin}_n(p \oplus p') = \text{lin}_n p \oplus \text{lin}_n p'$ and $\text{lin}_n(\text{id}_{\pi^*D} \otimes p) = \text{id}_{\pi^*D} \otimes \text{lin}_n p$ for a bundle $D \rightarrow X$.

The (i, j) th entry of the matrix $\text{lin}_n p$ acts on j th summand and takes it to the i th summand. So lin is linear in z .

Define a sequence $p_r(z)$ inductively by: $p_0 = p, zp_{r+1}(z) = p_r(z) - p_r(0)$. Then we can write $\text{lin}_n(p) = (1 + N_1)(p \oplus 1)(1 + N_2)$ where N_i are nilpotent matrices. If N is nilpotent, then $1 + tN$ is nonsingular for $0 \leq t \leq 1$.

Proposition 2.5. $\text{lin}_n(p)$ and $p \oplus 1$ define isomorphic bundles on P .

$$\mathcal{V}(E_0, p, E_\infty) \oplus \mathcal{V}\left(\sum_{k=1}^n L^k \otimes E_0, 1, \sum_{k=1}^n L^k \otimes E_0\right) \cong \mathcal{V}\left(\sum_{k=0}^n L^k \otimes E_0, \text{lin}_n(p), E_\infty \oplus \sum_{k=1}^n L^k \otimes E_0\right)$$

Lemma 2.6. Let p be a polynomial clutching function of degree $\leq n$ for (E_0, E_∞) . Then

1. $\text{lin}_{n+1}(E_0, p, E_\infty) \cong \text{lin}_n(E_0, p, E_\infty) \oplus (\text{lin}_{n+1} \otimes E_0, 1, \text{lin}_{n+1} \otimes E^0)$.
2. $\text{lin}_{n+1}(L^{-1} \otimes E_0, zp, E_\infty) \cong \text{lin}_n(E_0, p, E_\infty) \oplus (L^{-1} \otimes E_0, z, E_0)$

Proposition 2.7. For any polynomial clutching function p for (E_0, E_∞) , we have the identity in $K(P)$

$$([E_0, p, E_\infty] - [E_0, 1, E_0])(LH - 1) = 0$$

If we take $E_0 = 1, p = z, E_\infty = L$ we get $(H - 1)(LH - 1) = 0$.

Suppose $T \in \text{End}(E)$ for some finite dimensional vector space E . Let S be a circle in $\mathbb{C}$ which does not contain any eigenvalue of T . Then

$$Q_T = \frac{1}{2\pi i} \int_S \frac{1}{zI - T} dz$$

is a projection operator in E which commutes with T . The decomposition

$$E = E_+ \oplus E_-$$

where $E_+ = QE, E_- = (1 - Q)E$ is invariant under T . So T can be decomposed as

$$T = T_+ \oplus T_-$$

T_+ has all eigenvalues inside S while T_- has all of its eigenvalues outside S .

Proof. $T_\lambda I$ is invertible $\forall z \in S^1$ and commutes with T , so does Q_T . So Q_T preserves the eigenspaces of T . We can prove the proposition by considering just one eigenvalue.

$|T - \lambda I|$ is invertible iff $\lambda \notin \sigma(T)$. So if eigenvalue $|\lambda| > 1$, then $T - \lambda I$ is invertible inside D^2 and if $|\lambda| < 1$, then $T - \lambda I$ is homomorphic outside D^2 with a possible pole at ∞ . By change of variables $z \mapsto 1/z$, we can see that $Q_T = I$. $\square$

We defined $p(z)$ as sections over S . We extend the linear gluing function l , over the disk bundle $P_0 \rightarrow X$ to make it a section in $\Gamma(\pi_0^* E_0, \pi_0^* E_\infty)$. While similarly $1/zl(z)$ is a section in $\Gamma(\pi_\infty^* E_0, \pi_\infty^* E_\infty)$. Over $X_\infty \subset P_\infty$ the latter restricts to a . The values of these sections off S no longer need to be isomorphism. (Atiyah) If we assert that $l(z)$ is an isomorphism outside S , we shall take this to include the statement that a is an isomorphism.

Proposition 2.8. *Let l be a linear clutching function for (E_0, E_∞) and define endomorphism Q^0, Q^∞ of E_0, E_∞ by putting*

$$Q_x^0 = \frac{1}{2\pi i} \int_{S_x} l^{-1} dl \quad Q_x^\infty = \frac{1}{2\pi i} \int_{S_x} dl l$$

Then Q^0, Q^∞ are projection operators and $pQ^0 = Q^\infty p$.

Write $E_+^i = Q^i E^i, E_-^i = (1 - Q^i) E^i, i = 0, \infty$, so that $E^i = E_+^i \oplus E_-^i$. Then p is compatible with these decompositions, so that $p = p_+ \oplus p_-$. Moreover, p_+ is an isomorphism outside S , and p_- is an isomorphism inside S

Proof. WLOG we can assume $X = \text{pt}$, so $L = \mathbb{C}$ and z is just a complex number. Since l is an isomorphism for $|z| = 1$, and the set of isomorphisms is an open subset of hom , we can find a real number α , with $\alpha > 1$ such that $l(\alpha) : E_0 \rightarrow E_\infty$ is an isomorphism. We can now identify E_0, E_∞ with this isomorphism.

Consider the Möbius map $w = \frac{1-\alpha z}{z-\alpha}$ which preserves the unit circle and its inside. This gives $l(z) = \frac{w-T}{w+\alpha}$ where $T \in \text{End}(E_0)$. We get $Q^0 = Q^\infty = Q$. We can infer from this that l_+ and l_- are isomorphisms. $\square$

Corollary 2.9. *Let l be a linear gluing function write*

$$l_+ = a_+ z + b_+, \quad l_- = a_- z + b_-$$

Then, if $l(t) = l_+(t) \oplus l_-(t)$, where

$$l_+(t) = a_+ z + t b_+, \quad l_-(t) = t a_- z + b_-, \quad 0 \leq t \leq 1,$$

we obtain a homotopy of linear clutching functions connecting p with $a_+ z \oplus b_-$. Thus

$$(E^0, l, E^\infty) \cong (E_+^0, z, L \otimes E_+^0) \oplus (E_-^0, 1, E_-^0)$$

We go back to a polynomial clutching function p of degree $\leq n$. Then $\text{lin}_n(p)$ is a linear clutching function for (V^0, V^∞) where $V^0 = \sum_{k=0}^n L^k \otimes E^0$ $V^\infty = E^\infty \oplus \sum_{k=1}^n L^k \otimes E^0$.

Hence it defines a decomposition $V^0 = V_+^0 \oplus V_-^0$. To express the dependence on p and n . we write

$$V_+^0 = \mathcal{V}_n(E_0, p, E_\infty)$$

This is a vector bundle on Z . If p_t is a homotopy of polynomial clutching functions of degree $\leq n$ then

$$\mathcal{V}_n(E^0, p_0, E^\infty) \cong \mathcal{V}_n(E^0, p_1, E^\infty)$$

This gives us

$$\begin{aligned} \mathcal{V}_{n+1}(E^0, p, E^\infty) &\cong \mathcal{V}_n(E^0, p, E^\infty), \\ \mathcal{V}_{n+1}(L^{-1} \otimes E^0, zp, E^\infty) &\cong \mathcal{V}_n(E^0, p, E^\infty) \oplus (L^{-1} \otimes E^0) \end{aligned}$$

and

$$\mathcal{V}_{n+1}(E^0, zp, L \otimes E^\infty) \cong L \otimes \mathcal{V}_n(E^0, p, E^\infty) \oplus E^0$$

Combining this with the above corollary and 2.5, we obtain the following formula in $K(P)$:

$$\begin{aligned} [E^0, p, E^\infty] + \left\{ \sum_{k=1}^n [L^k \otimes E^0] \right\} [1] &= [v_n(E^0, p, E^\infty)] [H^{-1}] \\ + \left\{ \sum_{k=0}^n [L^k \otimes E^0] - [v_n(E^0, p, E^{\infty g})] \right\} [1]. \end{aligned}$$

and hence the formula

$$[E^0, p, E^\infty] = [\mathcal{V}_n(E^0, p, E^\infty)] ([H^{-1}] - [1]) + [E^0] [1].$$

This shows that $[\mathcal{V}_n(E^0, p, E^\infty)] \in K(X)$ completely determines $[E_0, p, E_\infty] \in K(P)$.

Let t be an indeterminant over the ring $K(X)$. Then the map have a $K(X)$ algebra homomorphism

$$\begin{aligned} \mu : \frac{K(X)[t]}{(t-1)(Lt-1)} &\rightarrow K(P) \\ t &\mapsto [H] \end{aligned}$$

To show μ is an isomorphism, we construct an inverse.

First suppose that f is a clutching function for (E_0, E_∞) . let f_n be the sequence of Cesaro means of its Fourier series, and put $p_n = z^n f_n$. Then for large n , p_n is a polynomial clutching function (of degree $\leq 2n$) for $(E_0, L^n \otimes E_\infty)$. We define

$$\nu_n(f) := [\mathcal{V}_{2n}(E_0, p_n, L^n \otimes E_\infty)](t^{n-1} - t^n) + [E_0]t^n$$

For large n , we have a homotopy between p_{n+1} and zp_n . This gives us

$$\mathcal{V}_{2n+2}(E_0, p_{n+1}, L^{n+1} \otimes E_\infty) \cong L \otimes \mathcal{V}_{2n}(E_0, p_n, L^n \otimes E_\infty) \oplus E_0$$

Henc $\nu_{n+1}(f) = \nu_n(f)$. Thus $\nu_n(f)$ is independent of n for large n . We write it as $\nu(f)$.

If g is close to f and n is large, then $f_n \simeq_n$ as polynomial clutching functions of degree $\leq 2n$ and hence

$$\nu(f) = \nu_n(f) = \nu_n(g) = \nu(g)$$

Thus $\nu(f)$ is a locally constant function of f and hence depends only on the homotopy class of f . However if E is any bundle on P and f a clutching function defining E , we define $\nu(E) = \nu(f)$. This is well-defined, additive, and depends only on isomorphism class of E . We have a group homomorphism

$$\nu : K(P) \rightarrow \frac{K(X)[t]}{(t-1)(Lt-1)}$$

$K(P)$ is additively generated by elements of the form $[E]$, so $\mu\nu(E) = E$ proves $\mu\nu = \text{id}$. Similarly $\nu\mu(t^n) = t^n$ for all $n \geq 0$. So $\nu\mu = \text{id}$.

3 K-theory of $\mathbb{C}P^n$

Theorem 3.1. *Let $L_1, \dots, L_n$ be line bundles over X and let H be the standard bundle $(\mathcal{O}(-1))$ over $P(L_1 \oplus \dots \oplus L_n)$. Then, the map*

$$\frac{K(X)(t)}{\prod_{i=1}^n (t - [L_i^*])} \xrightarrow[t \mapsto [H]]{\cong} K(P(L_1 \oplus \dots \oplus L_n))$$

induces an isomorphism of $K(X)$ -modules.

Proof. Let $p : P(E) \rightarrow X$ be a vector bundle. If L is a line-bundle over X . Then we have an isomorphism between

$$\begin{aligned} p(E) &\xrightarrow{\cong} p(E \otimes L) \\ v &\mapsto v \otimes l \end{aligned}$$

for some $l \in L$. This map takes the standard hyperplane bundle H over $P(E)$ to the bundle $H \otimes p^*L$ over $p(E \otimes L)$.

Multiplying each bundle L_i by L_1^{-1} replaces $[H]$ by $[H][L_1]$ and gives an isomorphism between $P(L_1 \oplus \dots \oplus L_n) \simeq P(1 \oplus \dots \oplus P_n)$. So WLOG, we can assume that $L_1 = 1$.

We will argue by induction on n . The case $n = 0$ being trivial. For brevity write $P_m = P(L_1 \oplus \dots \oplus L_m)$ for $1 \leq m \leq n$. We have inclusions $X \cong P_1 \rightarrow P_2 \rightarrow \dots \rightarrow P_n$. If H_m denotes the hyperplane bundle over P_m , then $H_m|_{P_{m-1}} \cong H_{m-1}$.

We have a commutative diagram

$$\begin{array}{ccccc} P_{n-1} & \xrightarrow{s} & P(H_{n-1}^* + 1) & \xleftarrow{\text{red}} & q^*(H_n) \cong G \\ \downarrow & & \downarrow q & & \downarrow \text{red} \\ X \approx P_1 & \longrightarrow & P_n & \xleftarrow{\text{red}} & H_n \end{array}$$

The map $q : P(H^* + 1) \rightarrow P_n$ is defined by sending $[v_1 : \dots : v_n]_{[w_1 : \dots : w_{n-1}]} \mapsto [v_1 : \dots : v_n]$ where $[v_1 : \dots : v_{n-1}]$ is a multiple (possibly 0) of $[w_1 : \dots : w_{n-1}]$ and s the zero section. This induces a homeomorphism

$$\frac{P^*(H_{n-1} \oplus 1)}{s(P_{n-1})} \cong \frac{P_n}{i(P_1)}$$

$q^*(H_n) = G$ is the standard hyperplane bundle over $P(H^* + 1)$.

From the Periodicity theorem we know that

$$KP(1 \oplus H_{n-1}^*) \approx \frac{KP_{n-1}[G]}{(G-1)(G-H_{n-1}^*)} \approx KP_{n-1} \oplus K(P(H_{n-1}^*), sP_{n-1})$$

Since $s^*[G] = [1]$ it follows from the long exact sequence of cohomology groups that $k(P(H_{n-1}^* + 1), s(P_{n-1}))$ is the submodule generated by $[G] - 1$ and that on this submodule multiplication by $[G]$ and $[H_{n-1}]$ coincide.

Translating to the pair (P_n, P_1) , we get that $K(P_n, P_1)$ is a free KP_{n-1} module with generator $[H] - 1$ and

$$H_{n-1} \cdot x = [H_n] \cdot x \quad \forall x \in K(P_n, P_1)$$

We now use the inductive assumption that the proposition holds for $n - 1$ in place of n and thus may assume the isomorphism

$$KP_{n-1} = \frac{KX[H_{n-1}]}{\prod([L_i][H_{n-1}] - 1)}$$

We have a commutative diagram

$$\begin{array}{ccc} \frac{KX(t)}{\prod([L_i]t-1)} & \xrightarrow{t \mapsto H_{n-1}^*} & KP_{n-1} \\ t-1 \downarrow & & \downarrow H-1 \\ \frac{(t-1)KX(t)}{(t-1)\prod([L_i]t-1)} & \xrightarrow{t \mapsto H} & K(P_n, P_1) \end{array}$$

X is a retract of P_1 and we have a decomposition $KP_n \simeq KX \oplus K(P_n, X)$. Complementing the bottom row by the term KX on both sides gives the KX algebra isomorphism

$$\frac{KX[H]}{\prod(H - [L_i^{-1}])} \simeq KP_n$$

completing the induction step. □

4

We describe an alternating description of $K(X, A)$ which has particular relevance for products.

If $n \geq 1$ we define $C_n(X, A)$ to be a category as follows: An object of $C_n(X, A)$ is a collection of $E_n, E_{n-1}, \dots, E_0$ of bundles over X together with maps $\alpha_i : E_i|_A \rightarrow E_{i-1}|_A$ such that

$$0 \rightarrow E_n|_A \xrightarrow{\alpha_n} E_{n-1}|_A \cdots \xrightarrow{\alpha_1} E_0|_A \rightarrow 0$$

is exact. The morphism $\phi : E \rightarrow F$ where $E = (E_i, \alpha_i), F = (F_i, \beta_i)$ are collection of maps ϕ_i such that $\beta_i \phi_i = \phi_{i-1} \alpha_i$.

An elementary sequence in $C_n(X, A)$ is a sequence of the form $0, 0, \dots, E_p, E_{p-1}, 0, \dots, 0$ where $E_p = E_{p-1}$.

We define $E \simeq F$ if for some set of *elementary objects* $Q_1, \dots, Q_k, P_1, \dots, P_l$ we have

$$E \oplus Q_1 \oplus \cdots \oplus Q_k \simeq F \oplus P_1 \oplus \cdots \oplus P_l$$

The set of such equivalence classes is denoted by $\mathcal{L}_n(X, A)$. This is a semigroup for each n . There is a natural inclusion $C_n(X, A) \subset C_{n+1}(X, A)$ which induces homomorphism $\mathcal{L}_n(X, A) \rightarrow \mathcal{L}_{n+1}(X, A)$. We denote by $C_\infty(X, A)$ the union of all C_n and by $\mathcal{L}_\infty(X, A)$ the direct limit $\mathcal{L}_n$.

Theorem 4.1. *For all $n \geq 1$, the maps*

$$\mathcal{L}_n(X, A) \rightarrow \mathcal{L}_{n+1}(X, A)$$

are isomorphisms and

$$\mathcal{L}_n(X, A) \simeq K(X, A)$$

Definition 4.2. *An Euler characteristic χ_n for $\mathcal{L}_n$ is a transformation of functors*

$$\chi_n : \mathcal{L}_n(X, A) \rightarrow K(X, A)$$

such that whenever $A = \emptyset, \chi(E_n, E_{n-1}, \dots, E_0) = \sum (-1)^i [E_i]$.

Lemma 4.3. *Let $A \subseteq X$ and let E, F be bundles over X . Let $\phi : E|_A \rightarrow F|_A, \psi : E \rightarrow F$ be monomorphisms (isomorphisms) and assume $\psi|_A \simeq \phi$. Then ϕ extends to X as a monomorphism (isomorphism).*

Lemma 4.4. *If $A = \text{pt}$, then*

$$0 \rightarrow \mathcal{L}_1(X, \text{pt}) \rightarrow \mathcal{L}_1(X) \rightarrow \mathcal{L}_1(\text{pt})$$

is exact. Thus if χ_1 is an Euler characteristic for $\mathcal{L}_1$,

$$\chi_1 : \mathcal{L}_1(X, \text{pt}) \xrightarrow{\sim} K(X, \text{pt})$$

is an isomorphism.

Proof. If (E_1, E_0) represents an element of $\mathcal{L}_1(X)$ whose image in $\mathcal{L}_1(\text{pt})$ is 0, E_1 and E_0 have the same dimension over A . Thus there is an isomorphism $\phi : E_1|_A \rightarrow E_0|_A$.

If (E_1, E_0, ϕ) has image zero in $\mathcal{L}_1(X)$, there is a trivial P and an isomorphism $\psi : E_1 \oplus P \cong E_0 \oplus P$. $\psi(\phi \oplus 1)^{-1}$ is an automorphism of $E_0 \oplus P|_A$. Since A is a point any such automorphism must be homotopic to the identity and hence it extends to $\alpha : E_0 \oplus P \cong E_0 \oplus P$. Thus we have a commuting diagram:

$$\begin{array}{ccc} E_1 \oplus P|_A & \xrightarrow{\phi \oplus 1} & E_0 \oplus P|_A \\ \downarrow \psi|_A & & \downarrow \alpha|_A \\ E_0 \oplus P|_A & \xrightarrow{1} & E_0 \oplus P|_A \end{array}$$

Thus (E_1, E_0, ϕ) represents 0 in $\mathcal{L}_1(X, A)$. □

Lemma 4.5.

$$\mathcal{L}_1(X/A, A/A) \xrightarrow{\sim} \mathcal{L}_1(X, A)$$

is an isomorphism for all (X, A) . Thus if χ_1 is an Euler characteristic,

$$\chi_1 : \mathcal{L}_1(X, A) \cong K(X, A)$$

is an isomorphism for all (X, A)

Lemma 4.6. If χ_1, χ'_1 are two Euler characteristics for $\mathcal{L}_1$, then $\chi_1 = \chi'_1$.

Proof. $\chi'_1 \chi_1^{-1}$ is a transformation of functors from K to itself which is the identity on each $K(X)$. Since $K(X, A) = \tilde{K}(X/A)$ is injected into $K(X/A)$, it is the identity on all $K(X, A)$. □

Theorem 4.7. There exists an Euler characteristic χ_1 for $\mathcal{L}_1$

Proof. Suppose $[E_1, E_0, \alpha] \in \mathcal{L}_1(X, A)$. Let X_0, X_1 be two copies of X and let $Y = X_0 \cup_A X_1$ be the space which results from identifying corresponding points of A . Then $[E_1, \alpha, E_0] \in K(Y)$. Let $r_i : Y \rightarrow X_i$ denote the retractions for $i = 1, 2$. Then

$$K(Y) = K(Y, X_1) \oplus K(X_1)$$

The map $(X_0, A) \rightarrow (Y, X_1)$ induces an isomorphism

$$K(Y, X_1) \rightarrow K(X_0, A)$$

Let $\chi_1(E_1, E_0, \alpha)$ be the image of the component of $[E_1, \alpha, E_0]$ which lies in $K(Y, X_1)$.

If $A = \emptyset$, then

$$\chi(E_1, E_0, \alpha) = [E_0] - [E_1]$$

This definition is independent of the choices made.

Two bundles
each over two
copies of X

$$\begin{array}{ccc} \mathcal{L}_1(X_0, A) & \xrightarrow{\sim} & \mathcal{L}_1(Y, X_1) \\ \downarrow & & \downarrow \\ K(X_0, A) & \xrightarrow{\sim} & K(Y, X_1) \end{array} \quad \begin{array}{c} [E_1, \alpha, E_0] \\ \downarrow \\ \chi_1([E_1, \alpha, E_0]) = \\ [E_1, \alpha, E_0] |_{K(Y, X_1)} \\ \stackrel{A=\emptyset}{=} [E_0] - [E_1] \end{array}$$

□

Corollary 4.8. *The class of (E_1, E_0, α) in $\mathcal{L}_1(X, A)$ only depends on the homotopy class of α .*

Proof. Let $Y = X \times [0, 1]$ and $B = A \times [0, 1]$. Then if α_t is a homotopy with $\alpha_0 = \alpha$, α_t defines $\beta : \pi^*(E_1)|_B \cong \pi^*(E_0)|_B$. Let $i_j : (X, A) \rightarrow (X \times [j], A \times [j])$. From the commuting diagram, every vertical map is an isomorphism and $i_0^*(i_1^*)^{-1}$ is the identity, we have $(E_1, E_0, \alpha_0) = (E_1, E_0, \alpha_1)$.

$$\begin{array}{ccccc}
 (E_1, E_0, \alpha_0) & & & & (E_1, E_0, \alpha_1) \\
 \\
 \mathcal{L}_1(X, A) & \xleftarrow{i_0^*} & \mathcal{L}_1(Y, B) & \xrightarrow{i_1^*} & \mathcal{L}_1(X, A) & (E_1, E_0, \alpha_t) \approx i_t^*(\pi^*(E_1), \pi^*(E_0), \beta) \\
 \downarrow \approx \chi_1 & & \downarrow \approx \chi_1 & & \downarrow \approx \chi_1 \\
 K(X, A) & \xleftarrow{i_0^*} & K(Y, B) & \xrightarrow{i_1^*} & K(X, A) \\
 & \nwarrow & & \nearrow & \\
 [E_1, \alpha_0, E_0] & & i_0^*(i_1^*)^{-1} = \text{id} & & [E_1, \alpha_1, E_0]
 \end{array}$$

□

Lemma 4.9. *The map $\mathcal{L}_n(X, A) \rightarrow \mathcal{L}_{n+1}(X, A)$ is onto for $n \geq 1$*

Proof. If $(E_{n+1}, \dots, E_0; \alpha_{n+1}, \dots, \alpha_1)$ represents an element of $\mathcal{L}_{n+1}(X, A)$, so does

$$(E_{n+1}, E_n \oplus E_{n+1}, E_{n-1} \oplus E_{n+1}, E_{n-2}, \dots, E_0; \alpha_{n+1}, \alpha_n \oplus 1, \dots, \alpha_1).$$

The two maps

$$\alpha_{n+1} \oplus 0 : E_{n+1} \rightarrow E_n \oplus E_{n+1}$$

and

$$0 \oplus 1 : E_{n+1} \rightarrow E_n \oplus E_{n+1}$$

are (linearly) homotopic as monomorphisms. $0 \oplus 1$ extends to X , and thus $\alpha_{n+1} \oplus 0$ extends to a monomorphism $\beta : E_{n+1} \rightarrow E_n \oplus E_{n+1}$ on all of X . Thus we can write $E_n \oplus E_{n+1}$ as $\beta(E_{n+1}) \oplus Q$. Then we see that, if $\gamma : Q \rightarrow E_{n-1} \oplus E_{n+1}$ is the resulting map,

$$(E_{n+1}, \dots, E_0; \alpha_{n+1}, \dots, \alpha_1) \sim (0, 0, E_{n-1} \oplus E_{n+1}, \dots, E_0; 0, \gamma, \dots, \alpha_1)$$

. Thus $\mathcal{L}_n(X, A) \rightarrow \mathcal{L}_{n+1}(X, A)$ is onto.

□

Lemma 4.10. *The map $\mathcal{L}_n(X, A) \rightarrow \mathcal{L}_{n+1}(X, A)$ is an isomorphism for all $n \geq 1$.*

Proof. It suffices to produce a map $\mathcal{L}_{n+1} \rightarrow \mathcal{L}_1$ which is a left inverse of the map $\mathcal{L}_1 \rightarrow \mathcal{L}_{n+1}$.

$$\begin{array}{ccccc} \mathcal{L}_1 & \hookrightarrow & \mathcal{L}_{n-1} & \hookrightarrow & \mathcal{L}_n \\ & & & & \\ & \swarrow & & \searrow & \\ & F & & (0, E) & \\ & \nwarrow & \nearrow & & \\ & (0, F) & & & \end{array}$$

Here

$$E = (E_n, \dots, E_0; \alpha_n, \dots, \alpha_1) \in \mathcal{L}_n \text{ and } F = (\sum E_{2i+1}, \sum E_{2i}, \sum \alpha_{2i+1} + \sum \alpha'_{2i}) \in \mathcal{L}_1$$

where α' is the Hermitian adjoint of α with respect to a choice of Hermitian metric on E_i . $\square$

Corollary 4.11. *For each n there exists exactly one Euler characteristic $\chi_n : \mathcal{L}_n(X, A) \rightarrow K(X, A)$ and it is always an isomorphism. Thus there exists*

$$\chi : \mathcal{L}_\infty(X, A) \rightarrow K(X, A)$$

isomorphically.

We want to construct pairings,

$$\mathcal{L}_n(X, Y) \otimes \mathcal{L}_m(X', Y') \rightarrow \mathcal{L}_{n+m}((X, Y) \times (X', Y'))$$

compatible with the pairings

$$K(X, Y) \otimes K(X', Y') \rightarrow ((X, Y) \times (X', Y'))$$

Consider complexes of vector bundles, i.e., sequences

$$0 \rightarrow E_n \xrightarrow{\sigma_n} E_{n-1} \xrightarrow{\sigma_{n-1}} \dots \rightarrow E_0 \rightarrow 0$$

where $\sigma_i \sigma_{i+1} = 0$ for all i .

Proposition 4.12. *Let $E_0, \dots, E_n$ be vector bundles on X and let $\sigma_i : E_i|_Y \rightarrow E_{i-1}|_Y$ be such that*

$$0 \rightarrow E_n \xrightarrow{\sigma_n} E_{n-1} \xrightarrow{\sigma_{n-1}} \dots \rightarrow E_0 \rightarrow 0 \quad (1)$$

is exact on Y . Then the σ_i can be extended to whole of X such that it forms a complex, i.e., $\tilde{\sigma}_i \tilde{\sigma}_{i+1} = 0$.

Proof. We shall show that there is some open neighborhood U of Y in X and an extension τ_i of σ_i to U for all i such that (1) is exact on U . The extension to the whole of X is then achieved by replacing τ_i with $\rho \cdot \tau_i$ where ρ is a continuous function on X such that $\rho = 1$ on Y and $\text{supp } \rho \subset U$.

Suppose that on some closed neighborhood of U_i of Y in X , we could extend $\sigma_1, \dots, \sigma_i$ to $\tau_1, \dots, \tau_i$ such that on U_i

$$E_i \xrightarrow{\tau_i} E_{i-1} \rightarrow \dots \rightarrow E_0 \rightarrow 0$$

is exact.

$K_i = \ker \tau_i$ and σ_{i+1} defines a section of $\text{hom}(E_{i+1}, K_i)$ and we have

$$\begin{array}{ccc} \sigma_{i+1} : E_{i+1} & \xrightarrow{\quad} & E_i \\ & \searrow & \nearrow \\ & K_i & \end{array}$$

This section can be extended to a neighborhood of Y in U_i and thus σ_{i+1} can be extended to τ_{i+1} on this neighborhood. Fill in the remaining details and use induction. $\square$

We introduce the set $\mathcal{V}_n(X, Y)$ of complexes of length n on X which are acyclic(exact) on Y . We say that two such complexes are homotopic if they are isomorphic to the restrictions to $X \times \{0\}$ and to $X \times \{1\}$ of an element in $\mathcal{V}_n(X \times I, Y \times I)$.

same as $\mathcal{L}_n$
without $\sim$

$$\begin{array}{ccc} & \mathcal{V}_n(X \times I, Y \times I) & \\ & \swarrow \quad \searrow & \\ \mathcal{V}_n(X \times 0, Y \times 0) & & \mathcal{V}_n(X \times 1, Y \times 1) \\ \downarrow \cong & & \downarrow \cong \\ \mathcal{V}_n^0(X, Y) & \xrightarrow{\quad \cong \quad} & \mathcal{V}_n^1(X, Y) \end{array}$$

There is a natural map

$$\Phi : \mathcal{V}_n(X, Y) \rightarrow \mathcal{L}_n(X, Y)$$

given by restriction of homomorphisms.

Lemma 4.13. Φ induces a bijection of homotopy classes.

Proof. The last lemma shows that Φ is surjective.

To show injectivity, we show that any complex over $X \times 0 \sqcup X \times 1 \sqcup Y \times I$ which is acyclic over $Y \times I$ can be extended to whole of $X \times I$. Apply the previous lemma to $X \times [0, 1/4]$ and $X \times [3/4, 1]$ and to the pair $X \times [1/4, 3/4], Y \times [1/4, 3/4] \cup V \times 1/4 \cup V \times 3/4$ and multiply with a ρ as before with suitable support. Here V is a suitable neighborhood around Y over which the complexes are still exact. $\square$

If $E \in \mathcal{V}_n(X, Y), F \in \mathcal{L}_m(X', Y')$, then $E \otimes F$ is a complex on $X \times X'$ which is acyclic on $X \times Y' \cup Y \times X'$. Thus we have a natural pairing

$$\mathcal{V}_n(X, Y) \otimes \mathcal{V}_m(X', Y') \rightarrow \mathcal{V}_{m+n}((X, Y) \times (X', Y'))$$

which is compatible with homotopies. Thus by means of Φ , it induces a pairing

$$\mathcal{L}_n(X, Y) \otimes \mathcal{L}_m(X', Y') \rightarrow \mathcal{L}_{m+n}((X, Y) \times (X', Y'))$$

Lemma 4.14. For any two classes $x \in \mathcal{L}_n(X, Y), x' \in \mathcal{L}_m(X', Y')$

$$\chi(x \otimes x') = \chi(x)\chi(x')$$

Proof. comments: The earlier pairing on K was obtained from

$$\tilde{K}(X/A) \otimes \tilde{K}(Y/B) \rightarrow \tilde{K}((X/A) \wedge (Y/B))$$

The formula holds true when $Y = Y' = \varphi$. However the above mentioned pairing on K was the only natural pairing compatible with the pairings defined for the case $Y = Y' = \varnothing$. $\square$

With this lemma we have a very convenient description of the relative product. We want to give a new construction for the generator of $\tilde{K}(S^{2n})$.

Let $\wedge^*(V)$ denote the exterior algebra. We can regrad this in a natrual way as a complex of vector bundles over V . Thus we put $E_i = V \times \wedge^i(V)$ and define

$$V \times \wedge^i(V) \rightarrow V \times \wedge^{i+1}(V)$$

by

$$(v, w) \rightarrow (v, v \wedge w)$$

If $\dim v = 1$, then the complex has just one map and this is an isomorphism for $v \neq 0$. We know relative K -groups can be defined via triples $[E_0, E_1; \sigma]$ where $\sigma : E_0 \rightarrow E_1$ is an isomorphism on $V - 0$ and by 4.12 this can be extended to all of V .

Now consider the bundles $\wedge^0 \mathbb{C}$ and $\wedge^1 \mathbb{C}$ over $B(V)$ with an induced metric, $[\wedge^0 \mathbb{C}, \wedge^1 \mathbb{C}; \sigma]$, identifying them over $S(V)$. This yields the "reduced" Hopf bundle which is the dual of H^* . Thus we have a generator of $K(B(V), S(V)) \approx \tilde{K}(S^2)$ and a sign -1 .

Since $\wedge^*(V \oplus W) \cong \wedge^*(V) \otimes \wedge^*(W)$, it follows that for any V , $\wedge^*(V)$ defines a complex over V acyclic on $V - 0$ and this gives a generator for $\tilde{K}(S^{2n})$ except for a sign $(-1)^n$.

For vector spaces V, W we have

$$\wedge^*(V \oplus W) \cong \wedge^*(V) \otimes \wedge^*(W)$$

So for any V , $\wedge^*(V)$ defines a complex over V acyclic on $V - 0$. So we have a generator and a sign of $(-1)^n$ for $K(B(V), S(V)) \approx \tilde{K}(S^{2n})$.

More generally this construction applies to a vector bundle V over a space X . Define the *Thom Space* X^V as the one point compactification of V or equivalently $B(V)/S(V)$. Then $K(B(V), S(V)) \cong \tilde{K}(X^V)$ and the exterior algebra defines a generator, call it λ_V

$$K(B(V), S(V)) \cong \tilde{K}(X^V) \cong \mathbb{Z}(\lambda_V)$$

where λ_V satisfies:

1. λ_V restricts to a generator of $\tilde{K}(P^V)$ for each $P \in X$.
2. $\lambda_{V \oplus W} = \lambda_V \cdot \lambda_W$, where this product comes from $\tilde{K}(X^V) \times \tilde{K}(X^W) \rightarrow \tilde{K}(X^{V \oplus W})$ which in-turn comes from $\mathcal{V}(B(V), S(V)) \times \mathcal{V}(B(W), S(W)) \rightarrow \mathcal{V}((B(V), S(V)) \times (B(W), S(W)))$.

We carry out a similar discussion for projective spaces.

If V is a vector bundle, let $P = P(V \oplus 1)$ and let H be the standard line bundle over P . We have a monomorphism

$$H^* \rightarrow \pi^*(V \oplus 1)$$

when $\pi : P \rightarrow X$ is the projection. Hence tensoring with H

$$H^* \otimes H \rightarrow \pi^*(V \oplus 1) \otimes H$$

$$1 \rightarrow H \otimes \pi^*(V \oplus 1)$$

we get a section of $H \otimes \pi^*(V \oplus 1)$. Projecting onto the first factor gives a natural section

$$s \in \Gamma(H \otimes \pi^*V)$$

i.e., we have a commuting diagram

$$\begin{array}{ccc} H \otimes \pi^*V & \longrightarrow & H \otimes \pi^*W \\ \downarrow & \circlearrowleft & \downarrow \\ P(V) & \longrightarrow & P(W) \end{array}$$

Consider the exterior algebra $\wedge^*(H \otimes \pi^*(V))$. Each component is a vector bundle over P and exterior multiplication by s gives us a complex of vector bundles acyclic outside the subspace where $s = 0$. The subspace where $s = 0$ is the image of the X under the section $X \rightarrow P$.

$$\begin{array}{ccc} & H \otimes \pi^*(V) & \\ & \uparrow s & \\ P(V \oplus 1) & \longleftrightarrow & P(V) \\ & \uparrow & \\ & X & \end{array}$$

H restricted to $P(V \oplus 1) - P(V)$ is just the trivial bundle 1.

$$\begin{array}{ccccc} V \oplus 1 & \longleftarrow & H & \longleftarrow & \text{res}(H) \otimes \pi^*(V) \simeq \pi^*(V) \\ \downarrow & & \downarrow & & \downarrow \\ X & \longleftarrow & P(V \oplus 1) & \longleftarrow & P(V)^c \end{array}$$

We get $\lambda_{\pi^*(V)}$ which we denote as λ_V . The image of λ_V under the map

check

$$\begin{aligned} \tilde{K}(X^V) &= K(P(V \oplus 1), PV) \rightarrow K(P(V \oplus 1)) \\ \lambda_V &\mapsto \sum (-1)^i [H]^i [\wedge^i V] \end{aligned}$$

$$\begin{array}{ccc} L_n(P(V \oplus 1), p(V)) & \longrightarrow & L_n(P(V \oplus 1)) \\ \downarrow & & \downarrow \\ K(P(V \oplus 1), p(V)) & \longrightarrow & K(P(V \oplus 1)) \end{array}$$

All these results above can be extended to G bundles.

5 Thom isomorphism

E a vector bundle is said to be decomposable if it can be written as a sum of line bundles $E = \sum L_i$. Then we have determined that $K(PE)$ is a $K(X)$ algebra.

$$\tilde{K}(X \times S^1) \cong \tilde{K}(X \wedge S^1) \oplus \tilde{K}(X) \oplus \tilde{K}(S^1) = K^*(X)$$

If $\pi : X \times S^1 \rightarrow X$ is the projection, we have $P(E) \times S^1 = P(\pi^*E)$ and so

$$K^*(PE) \cong K(P(\pi^*E))$$

Thus considering $X \times S^1$ instead of X , we get

Proposition 5.1. *Let $E = \sum L_i$ be a decomposable vector bundle over X . then $K^*(PE)$ is a $K^*(X)$ algebra generated by $[H]$ subject to relation $\prod([L_i][H] - 1) = 0$.*

Remark 5.2. *This also extends to G -spaces giving $K_G^*(PE)$ as $K_G^*(X)$ algebra.*

The Thom space X^E can be identified with $P(E \oplus 1)/P(E)$ and we have the relation

$$\sum (-1)^i [H]^i [\wedge^i E] = \prod (1 - [L_i][H])$$

$$\begin{array}{ccccc} K(P(V \oplus 1), p(V)) & \longrightarrow & K(P(V \oplus 1)) & \longrightarrow & K(PE) \\ \parallel & & \parallel & & \parallel \\ <\lambda_V> & \longmapsto & \sum (-1)^i H^i [\wedge^i V] & \longmapsto & \frac{K^*(X)[H]}{\prod (1 - [L_i][H])} \end{array}$$

Since this element generates kernel of the map $K^*(P(E \oplus 1)) \rightarrow K^*(PE)$, we have

Proposition 5.3. *Let E be decomposable vector bundle over X . Then $\tilde{K}^*(X^E)$ is a free $K^*(X)$ module with λ_E as a generator.*

Remark 5.4. *This Proposition also holds for G -spaces.*

Corollary 5.5. *Let X be a G space such that $K_G^1(X) = 0$ and let E be a decomposable G vector bundle. Then, if $S(E)$ denotes the sphere bundle we have an exact sequence*

$$0 \rightarrow K_G^1(S(E)) \rightarrow K_G^0(X) \xrightarrow{\phi} K_G^0(X) \rightarrow K_G^0(SE) \rightarrow 0$$

where ϕ is multiplication by

$$\lambda_{-1}[E] = \sum (-1)^i \lambda^i[E]$$

Disk bundle contracts to origin and H restricts to 1 on $P(E^c)$.

Lemma 5.6. $K_G^1(pt.) = 0$

We consider X to be a point and G abelian group so that E is necessarily decomposable. We obtain the following corollary

Corollary 5.7. *Let G be an abelian group, E a G module. Then we have an exact sequence*

$$0 \rightarrow K_G^1(SE) \rightarrow R(G) \xrightarrow{\phi} R(G) \rightarrow K_G^0(SE) \rightarrow 0$$

where ϕ is multiplication by

$$\lambda_{-1}[E] = \sum (-1)^i \wedge^i [E]$$

Suppose G acts freely on SE over $X = pt.$, we have $K_G^*(SE) \cong K^*(SE/G)$

G need not be cyclic

Then we have

Corollary 5.8. *Let G be a cyclic group, E a G module with $S(E)$, G free bundle. Then we have an exact sequence*

$$0 \rightarrow K^1(SE/G) \rightarrow R(G) \xrightarrow{\phi} R(G) \rightarrow K^0(SE/G) \rightarrow 0$$

where ϕ is multiplication by $\lambda_{-1}[E]$.

5.9. We use this to calculate the K^* for $\mathbb{R}P^{n-1}$. Take $G = \mathbb{Z}_2$ and $E = \mathbb{C}^n$ with (-1) action. Then

$$S(E)/G = \mathbb{R}P^{2n-1}$$

The representation ring of $\mathbb{Z}_2$ is

$$R\mathbb{Z}_2 = \frac{\mathbb{Z}\rho}{\rho^2 - 1}$$

$$\lambda_{-1}[E] = (1 - \rho)^n$$

We have an exact sequence

$$0 \rightarrow K^1(\mathbb{R}P^{2n-1}) \rightarrow R(\mathbb{Z}/2) \xrightarrow{\phi} R\mathbb{Z}/2 \rightarrow K^0(\mathbb{R}P^{2n-1}) \rightarrow 0$$

where ϕ is multiplication by $\lambda_{-1}[E]$.

Set $\sigma = 1 - \rho$ so that $\sigma^2 = 2\sigma$, $\sigma^{j+1} = 2^j\sigma$ and $\lambda_{-1}[E] = (-\sigma)^n$ and in terms of basis $(1, \sigma)$ the map ϕ has a matrix

$$\begin{bmatrix} 0 & 0 \\ 2^{n-1} & 2^n \end{bmatrix}$$

. We thus obtain

$$K^0(\mathbb{R}P^{2n-1}) = K^0(*) \oplus \tilde{K}^0(\mathbb{R}P^{2n-1}) \quad \forall n > 0$$

On the other hand we have

$$K^1(\mathbb{R}P^{2n-1}) = (2 - \sigma) = (1 + \rho) \simeq \mathbb{Z} \quad \forall n > 0$$

For spaces of even dimension consider the commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & K^1(\mathbb{R}P^{2n+1}) & \longrightarrow & R\mathbb{Z}_2 & \xrightarrow{\sigma^{n+1}} & R\mathbb{Z}_2 \\ & & \downarrow \text{res} & & \downarrow \cdot \sigma & & \parallel_1 \\ 0 & \longrightarrow & K^1(\mathbb{R}P^{2n-1}) & \longrightarrow & R\mathbb{Z}_2 & \xrightarrow{\sigma^n} & R\mathbb{Z}_2 \end{array}$$

Multiplication by σ^n has the same kernel as multiplication by σ and σ^{n+1} . So the map res is 0 map. The quotient $\mathbb{R}P^m/\mathbb{R}P^{m-1}$ is a m and the vertical arrows are induced by the same maps as the horizontal maps, so we have the exact sequences

check why the dotted map is 0

$$\begin{array}{ccccccc} \mathbb{Z} = K^1(\mathbb{R}P^{2n+1}, \mathbb{R}P^{2n}) & \longrightarrow & \mathbb{Z} = K^1(\mathbb{R}P^{2n+1}) & \longrightarrow & K^1(\mathbb{R}P^{2n}) & \longrightarrow & K^0(\mathbb{R}P^{2n+1}, \mathbb{R}P^{2n}) = 0 \\ \downarrow & & \downarrow & \searrow \scriptstyle 0 & \downarrow & & \downarrow \\ 0 = K^1(\mathbb{R}P^{2n}, \mathbb{R}P^{2n-1}) & \longrightarrow & K^1(\mathbb{R}P^{2n}) & \longrightarrow & K^1(\mathbb{R}P^{2n-1}) = \mathbb{Z} & \longrightarrow & K^0(\mathbb{R}P^{2n}, \mathbb{R}P^{2n-1}) = \mathbb{Z} \end{array}$$

Thus

$$K^1(\mathbb{R}P^{2n}) = 0$$

Both $K^1(\mathbb{R}P^{2n+1})$ and $K^1(\mathbb{R}P^{2n+1}, \mathbb{R}P^{2n})$ are infinite cyclic groups. Thus we have an exact sequence

$$K^0(\mathbb{R}P^{2n+1}, \mathbb{R}P^{2n}) \rightarrow K^0(\mathbb{R}P^{2n+1}) \rightarrow K^0(\mathbb{R}P^{2n}) \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow 0$$

and the inclusion induces an isomorphism $K^0(\mathbb{R}P^{2n+1}) \rightarrow K^0(\mathbb{R}P^{2n})$, so that

$$K^0(\mathbb{R}P^{2n}) = K^0(*) \oplus \tilde{K}^0(\mathbb{R}P^{2n}) \simeq \mathbb{Z} \oplus \mathbb{Z}/2^n$$

Theorem 5.10. Let $\pi : B \rightarrow X$ be a map of compact spaces and let $\mu_1, \dots, \mu_n$ be homogeneous elements of $K^*(B)$. Let M^* be the free $\mathbb{Z}_2$ graded group generated by $\mu_1, \dots, \mu_n$. Suppose that every point $x \in X$ has a neighborhood U such that for all $V \subset U$, the natural map

compact necessary

$$K^*(V) \otimes M^* \rightarrow K^*(\pi^{-1}V)$$

is an isomorphism. Then for any $Y \subset X$, the map

$$K^*(X, Y) \otimes M^* \rightarrow K^*(B, \pi^{-1}(Y))$$

is an isomorphism.

Corollary 5.11. Let $\pi : E \rightarrow X$ be a vector bundle and let H be the usual line bundle over $P(E)$. Then $K^*(PE)$ is a free $K^*(X)$ module on the generators $[H]^i, 0 \leq i \leq n-1$ and $[H]$ satisfies the equation $\sum (-1)^i [H]^i [\wedge^i E] = 0$.

Corollary 5.12. If $\pi : E \rightarrow X$ is a vector bundle and if FE is the flag bundle of E with projection map $p : FE \rightarrow X$ then

$$p^* : K^*(X) \rightarrow K^*(FE)$$

is injective

Corollary 5.13 (The Splitting principle). If $E_1, \dots, E_n$ are vector bundles on X , then there exist a space F and a map $\pi : F \rightarrow X$ such that

1. $\pi^* : K^*(X) \rightarrow K^*(F)$ is injective.
2. Each $\pi^*(E_i)$ is a sum of line bundles.

This allows us to reduce problems to the decomposable case

As a corollary we have the theorem,

Theorem 5.14 (Thom Isomorphism theorem). *If $\pi : E \rightarrow X$ is a vector bundle*

$$\begin{aligned}\Phi : K^*(X) &\rightarrow \tilde{K}^*(X^E) \\ F &\mapsto \lambda_E \cdot F\end{aligned}$$